3/30/2022

Friday, March 25, 2022 10:46 AM

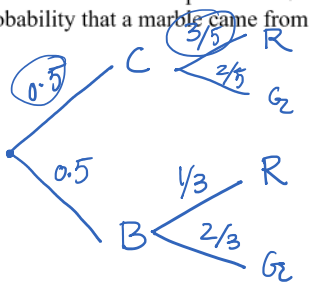
M168_Notes_7.6

7.6 Bayes' Theorem

$$P(R) = P(R \cap C) + P(R \cap B)$$

Example

We are to choose a marble from a cup or a bowl. We need to flip a coin to decide to choose from the cup or the bowl. The bowl contains 1 red and 2 green marbles. The cup contains 3 red and 2 green marbles. What is the probability that a marble came from the cup given that it is red?

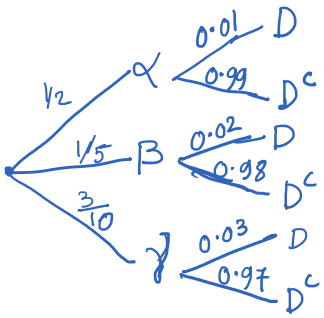


$$\boxed{P(C|R) = \frac{P(C \cap R)}{P(R)}}$$

$$= \frac{P(C \cap R)}{P(C \cap R) + P(B \cap R)} = \frac{\frac{1}{2} \times \frac{3}{5}}{\frac{1}{2} \times \frac{3}{5} + \frac{1}{2} \cdot \frac{1}{3}} =$$

Example

A company makes the components for a product at a central location. These components are shipped to three plants, Alpha, Beta, and Gamma, for assembly into a final product. The percentages of the product assembled by the three plants are, respectively, 50%, 20%, and 30%. The percentages of defective products coming from these three plants are, respectively, 1%, 2%, and 3%. Given a defective product, what is the probability it was assembled at plant Alpha?



$$P(\alpha|D) = \frac{P(\alpha \cap D)}{P(\alpha \cap D) + P(\beta \cap D) + P(\gamma \cap D)} = \frac{P(\alpha \cap D)}{P(D)}$$

$$= \frac{0.5 \times 0.01}{0.5 \times 0.01 + 0.2 \times 0.02 + 0.3 \times 0.03} =$$